

Impact Statement: Intelligent Complaint Classification in Financial Services

This project introduces a transformer-based text classification pipeline designed to automatically triage consumer banking complaints into operational departments. By replacing manual, error-prone routing with an open-source NLP architecture, this work directly addresses operational bottlenecks that delay dispute resolution in the financial sector.

Positive Societal & Economic Impacts

The primary benefit of this system is democratic access to efficient financial redress. In banking, delayed complaint resolution often disproportionately harms low-income or financially vulnerable consumers, who are most impacted by unresolved transaction errors, billing disputes, or hidden fees. By accelerating triage times, the system reduces resolution backlogs and ensures customer grievances are directed to the correct specialized department. Furthermore, deploying lightweight models like DistilBERT offers a cost-effective, open-source solution for small-to-medium financial institutions in under-resourced regions that lack the infrastructure for expensive proprietary customer service platforms.

Ethical Risks & Mitigation Strategies

The key ethical risk in automated triage is the systemic misclassification of high-severity complaints. In financial services, mislabeling an unauthorized transaction or fraud claim as a routine service inquiry can cause severe monetary loss, account lockout, and loss of consumer trust. To mitigate this, the architecture does not grant autonomous decision-making power to the model. Instead, it integrates three core safety safeguards:

1. **Confidence Thresholding:** Model outputs undergo Temperature Scaling calibration. Any prediction falling below a pre-set confidence threshold is automatically escalated to human agents.
2. **Urgency Flagging:** Minority, high-risk categories (e.g., fraud, account takeover) bypass general queues for immediate priority review.
3. **Human-in-the-Loop:** High-impact cases strictly require human verification before closing, preserving human agency and accountability.

Linguistic Fairness & Cross-Domain Representation

A critical limitation of language models trained on standard Western datasets (such as the CFPB corpus) is algorithmic bias against non-standard syntax, regional slang, and local idioms. When applied in broader contexts, standard English models risk higher error rates on non-Western phrasing, effectively penalizing customers based on how they communicate. To address this inequality, the system is explicitly evaluated against an independent corpus of African English

customer complaints to benchmark cross-domain generalization and ensure equitable performance across regional linguistic patterns.

Long-Term Governance & Accountability

Deploying AI in financial operations requires ongoing governance to prevent bias creep as consumer phrasing evolves. By enforcing rigorous privacy standards; specifically scrubbing personally identifiable information (PII) before processing and using transparent evaluation metrics like Macro F1-scores, this project demonstrates how financial institutions can innovate responsibly while respecting consumer privacy, dignity, and local context.